KODLAR

testbench.sv

```
module tb_adder_4bit;

    reg [3:0] A;
    reg [3:0] B;
    reg Cin;
    wire [3:0] Sum;
    wire Cout;

    adder_4bit uut (
        .A(A),
        .B(B),
        .Cin(Cin),
        .Sum(Sum),
        .Cout(Cout)
    );

    initial begin
        A = 0; B = 0; Cin = 0;
        #10;

        A = 4'b0011;
        B = 4'b0101;
        Cin = 0;
        #10;

        A = 4'b1111;
        B = 4'b0001;
        Cin = 0;
        #10;

        A = 4'b0111;
        B = 4'b0010;
        Cin = 1;
        #10;

        $finish;
    end

    initial begin
        $monitor("Zaman=%0t | A=%b, B=%b, Cin=%b | Toplam=%b, Elde=%b", $time, A, B, Cin, Sum, Cout);
    end

endmodule
```

design.sv

```
module adder_4bit (
    input [3:0] A,
    input [3:0] B,
    input Cin,
    output [3:0] Sum,
    output Cout
);

    assign {Cout, Sum} = A + B + Cin;

endmodule
```

ÇIKTI

```
Zaman=0 | A=0000, B=0000, Cin=0 | Toplam=0000, Elde=0
Zaman=10 | A=0011, B=0101, Cin=0 | Toplam=1000, Elde=0
Zaman=20 | A=1111, B=0001, Cin=0 | Toplam=0000, Elde=1
Zaman=30 | A=0111, B=0010, Cin=1 | Toplam=1010, Elde=0
testbench.sv:36: $finish called at 40 (1s)
Done
```